

(2 Marks)

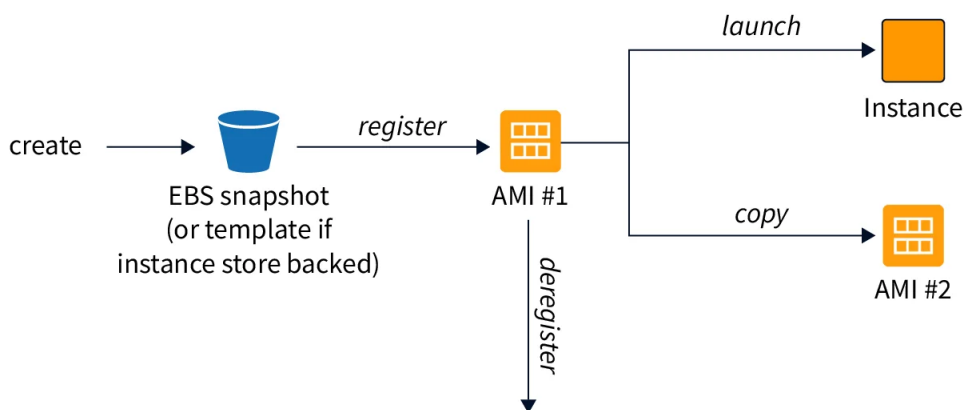
1. State the features of Amazon Web Services. (also Q.7)

- **On-Demand Computing:** Provides instant access to computing resources like servers, storage, and databases whenever needed.
- **Scalability:** Resources can be easily scaled up or down based on demand.
- **Pay-as-You-Go Pricing:** Users are charged only for the services they actually use.
- **High Availability:** Ensures services are available with minimal downtime using multiple data centers (regions and availability zones).
- **Global Infrastructure:** Has data centers across different countries for low-latency access worldwide.
- **Security:** Offers strong security features like encryption, IAM (Identity and Access Management), and firewalls.
- **Flexible Services:** Provides a wide range of services such as computing (EC2), storage (S3), databases, AI/ML, and networking.
- **Easy Deployment:** Applications can be deployed quickly using managed services and automation tools.
- **Reliability:** Built-in backup, replication, and disaster recovery options.
- **Elasticity:** Automatically adjusts resources based on workload changes.

2. What is Amazon Machine Image? Draw life cycle of AMI

Amazon Machine Image (AMI) is a pre-configured template used to launch virtual servers called EC2 instances in Amazon Web Services. It contains the operating system, application software, and necessary configuration settings required to run an instance. In simple terms, an AMI acts like a ready-made blueprint that allows users to quickly create and deploy virtual machines in the cloud without setting up everything from scratch.

Life cycle of AMI :



3. Explain the characteristics of cloud computing.

cloud storage service is a service that allows users to store data on remote servers over the internet instead of using local storage devices like hard drives or memory cards. It provides a flexible and efficient way to manage data with easy access, high security, and automatic backup features.

Key Points:

- Stores data on remote cloud servers instead of local devices
- Provides high security with encryption and access control
- Highly scalable (storage can be increased or decreased anytime)

- Offers automatic backup and data recovery options
- Accessible from anywhere using the internet
- Reduces cost by using pay-as-you-go pricing model
- Ensures high availability and durability of data
- Example: Amazon S3 used for storing files, images, videos, and backups.

4. Explain Amazon Glacier in short

Amazon Glacier (Amazon S3 Glacier) is a low-cost cloud storage service used for long-term data archiving and backup. It is mainly designed for storing data that is rarely accessed but must be preserved securely for a long time.

Key Points:

- Used for long-term backup and data archiving
- Very low-cost storage compared to standard storage services
- Data retrieval is slow (takes minutes to hours)
- Highly secure with encryption and access control
- Provides high durability and reliability
- Best for storing old records, logs, and compliance data
- Helps organizations reduce overall storage costs
- Integrated with Amazon S3 lifecycle policies for automatic archiving

5. What is CloudWatch?

Amazon CloudWatch is a monitoring and observability service that helps you monitor AWS resources, applications, and services in real time. It collects data such as metrics, logs, and events to understand system performance and detect problems early.

Key Points:

- Monitors AWS resources and applications in real time
- Collects metrics, logs, and event data automatically
- Helps in setting alarms for CPU usage, memory, and errors
- Sends notifications using alerts (via email or SMS)
- Supports visualization through dashboards
- Helps in troubleshooting and performance optimization
- Enables automatic actions when thresholds are crossed
- Improves reliability and system health monitoring

6. What is Amazon S3? State benefits of S3

Amazon S3 (Simple Storage Service) is a highly scalable, secure, and durable cloud storage service used to store and retrieve large amounts of data over the internet. It allows users to store any type of data such as files, images, videos, documents, backups, and application data in “buckets” and access them anytime from anywhere. Amazon S3 is designed to provide high performance, easy management, and flexible storage options for individuals and businesses of all sizes.

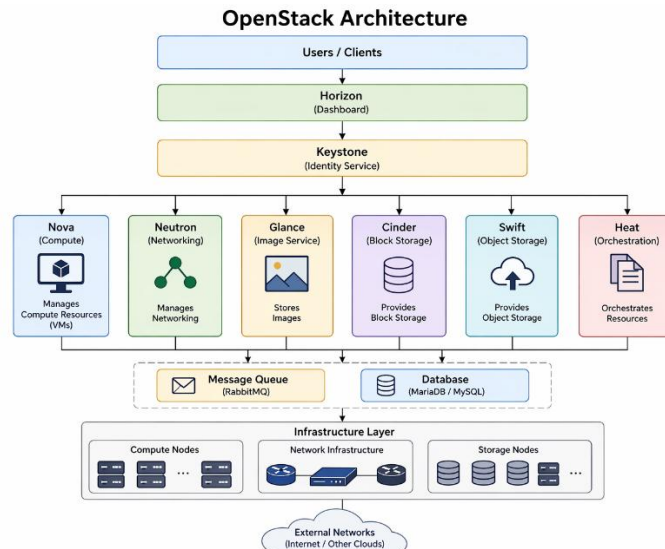
Benefits of Amazon S3:

- Highly scalable storage (store unlimited data)
- Secure with encryption and access control
- High durability and availability of data
- Cost-effective pay-as-you-go pricing model
- Easy data access from anywhere via internet
- Supports backup, restore, and disaster recovery
- Integrates with other AWS services

- Fast performance for data upload and download

(5 Marks)

7. Draw and Explain OpenStack Architecture. (also Q.12)



OpenStack is an **open-source cloud computing platform** used to build and manage **private and public clouds**. It allows organizations to control compute, storage, and networking resources through different integrated services. These services work together to create and manage virtual machines and cloud infrastructure efficiently.

OpenStack Architecture Explanation

- 1. Users / Clients** : Users or applications interact with the OpenStack cloud through web interfaces, APIs, or command line tools. They send requests to create virtual machines, manage storage, or configure networks.
- 2. OpenStack Horizon (Dashboard)** : Horizon is the graphical web interface of OpenStack. It allows users and administrators to manage cloud resources like virtual machines, networks, and storage easily.
- 3. OpenStack Keystone (Identity Service)** : Keystone handles authentication and authorization. It verifies user identity and controls access to different OpenStack services.
- 4. OpenStack Nova (Compute)** : Nova manages compute resources and is responsible for creating and running virtual machines in the cloud.
- 5. OpenStack Neutron (Networking)** : Neutron manages networking in OpenStack. It connects virtual machines and controls IP addresses, routers, and network security.
- 6. OpenStack Glance (Image Service)** : Glance stores and manages virtual machine images that are used to launch new instances.
- 7. OpenStack Cinder (Block Storage)** : Cinder provides persistent storage volumes that can be attached to virtual machines like a hard disk.
- 8. OpenStack Swift (Object Storage)** : Swift stores large amounts of unstructured data such as files, images, and backups.
- 9. OpenStack Heat (Orchestration)** : Heat automates the deployment of cloud infrastructure using templates.
- 10. Message Queue – RabbitMQ** : RabbitMQ enables communication between different OpenStack services.
- 11. Database – MariaDB / MySQL** : The database stores configuration and resource information used by

OpenStack services.

12. Infrastructure Layer : This layer includes compute nodes, network infrastructure, and storage nodes that provide the physical resources for the cloud.

8. Write a short note on: Benefits and challenges of Mobile Cloud Computing

(MCC) combines mobile devices with cloud computing to provide storage, processing power, and services through the internet. It allows mobile devices to perform complex tasks using cloud servers instead of relying only on their own hardware. **This improves the performance, scalability, and efficiency of mobile applications.**

Benefits

- **High Processing Power:** Cloud servers handle heavy computations, improving mobile performance.
- **Large Storage Capacity:** Users can store large amounts of data in the cloud instead of the device.
- **Anywhere Access:** Applications and data can be accessed anytime and from any location.
- **Reduced Device Limitations:** Mobile devices save battery and memory because processing happens in the cloud.
- **Cost Efficiency:** Users do not need expensive hardware since cloud resources provide computing power.
- **Automatic Updates and Backup:** Cloud services provide automatic data backup and software updates.

Challenges

- **Network Dependency:** Requires a stable internet connection to function properly.
- **Security and Privacy Issues:** Data stored in the cloud may be vulnerable to cyber threats.
- **Latency Problems:** Slow network speed can affect application performance.
- **Bandwidth Limitations:** Limited bandwidth can reduce efficiency when transferring large data.
- **Data Loss Risk:** Improper cloud management may lead to accidental data loss.
- **Service Availability Issues:** Cloud services may become unavailable due to server failures or maintenance.

9. Explain different types of security in cloud.

1. **Data Security :** Data security protects cloud data from unauthorized access, loss, or theft. It uses techniques such as encryption, data masking, and regular backups. This helps maintain confidentiality and integrity of data in the cloud.
2. **Network Security :** Network security protects cloud infrastructure and communication channels from cyber attacks. It uses tools like firewalls, VPNs, and intrusion detection systems. This ensures safe data transmission over the network.
3. **Application Security :** Application security focuses on protecting cloud-based applications from vulnerabilities and attacks. It includes secure coding practices, regular updates, and vulnerability testing. This helps prevent hackers from exploiting application weaknesses.
4. **Identity and Access Management (IAM) :** IAM ensures that only authorized users can access cloud resources. It uses authentication methods such as passwords, role-based access control, and multi-factor authentication. This reduces the risk of unauthorized access.
5. **Physical Security :** Physical security protects the hardware and data centers where cloud services are hosted. Measures include surveillance cameras, biometric access controls, and security personnel. This prevents physical damage or unauthorized entry.
6. **Endpoint Security :** Endpoint security protects devices like laptops, smartphones, and tablets that access cloud services. It uses antivirus software, device authentication, and security policies. This helps stop malware and other threats from user devices.
7. **Disaster Recovery and Backup Security :** This security ensures that data and services can be

restored after failures, cyber attacks, or natural disasters. Regular backups and recovery plans help maintain data availability. This minimizes downtime and data loss.

8. **Governance, Compliance, and Policy Controls** : This ensures that cloud systems follow organizational policies, legal regulations, and industry standards. It helps manage risks and enforce security rules. This ensures responsible and secure use of cloud resources.

10. What is Identity and Access Management in cloud security?

Identity and Access Management (IAM) is a security framework in cloud computing that manages digital identities and controls access to cloud resources. It ensures that only authenticated and authorized users can access specific systems, applications, and data in the cloud environment.

- **Authentication** : Authentication is the process of verifying the identity of a user or system before granting access to cloud services. Common authentication methods include passwords, security tokens, biometric verification, and multi-factor authentication (MFA) to enhance security.
- **Authorization** : Authorization determines the level of access or permissions granted to a user after successful authentication. It ensures that users can only perform actions or access resources that are permitted according to predefined policies.
- **Role-Based Access Control (RBAC)** : RBAC assigns permissions to users based on their roles within an organization, such as administrator, developer, or end-user. This approach simplifies access management and ensures users receive only the permissions necessary for their job functions.
- **User Identity Management** : IAM systems manage the lifecycle of user identities, including user creation, modification, and deletion. Proper identity management helps maintain control over who can access cloud services and prevents unauthorized access.
- **Policy and Access Control Management** : IAM allows organizations to define and enforce security policies that regulate access to cloud resources. These policies specify who can access what resources, under which conditions, and for what purpose.
- **Monitoring and Auditing** : IAM systems track and record user activities and access attempts within the cloud environment. Monitoring and auditing help detect suspicious behavior, ensure compliance with security standards, and support security investigations.
- **Security and Compliance Benefits** : By implementing IAM, organizations can significantly reduce the risk of unauthorized access, data breaches, and insider threats. It also helps organizations comply with regulatory requirements and maintain strong governance over cloud resources.